

Paul Rübsamen-von Döhren

PHD IN ENGINEERING · MATERIAL SCIENTIST

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Experience

PhD Researcher

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

EMPLOYMENT PERIOD

- Designed and developed an experimental setup for quantifying ice adhesion under controlled environmental conditions, relevant for marine and offshore applications
- Established measurement methodologies for surface-ice interaction phenomena, combining mechanical testing and environmental control systems
- Conducted experimental campaigns integrating materials testing, surface characterization, and high-resolution imaging techniques
- Contributed to multidisciplinary research within the GoNorth Arctic research initiative
- Experience in linking laboratory-scale experiments with real-world environmental conditions

Trondheim, Norway

Nov. 2020 - June 2026

Nov. 2020 - April 2024

Student Assistant / Research Collaborator

GEORG-AUGUST UNIVERSITÄT GÖTTINGEN & TOP STRAP

- Collaborated with Top Strap to optimize production processes for biodegradable strap materials, integrating laboratory findings into industrial-scale manufacturing
- Supported process optimization and material validation in collaboration with academic and industry engineers, advancing applied research into scalable manufacturing solutions

Göttingen, Germany

2017 - 2018

Intern / Undergraduate Researcher

AGILENT TECHNOLOGIES

- Gained hands-on experience with advanced instrumentation and quality-controlled laboratory environments in an industrial R&D setting
- Collaborated with researchers on sustainable material development with focus on environmentally responsible alternatives

Middelburg, Netherlands

Aug. 2017- Sep. 2017

Education

PhD in Engineering

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

- Dissertation: "Experimental Investigation of Microstructure and Adhesion of Sea-Spray Ice"
- Advanced coursework: Particle design and crystallization, ice mechanics, and nanomechanics (37 ECTS credits).

Trondheim, Norway

Nov. 2020 - June 2026

M.Sc. in Material Science

GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN

- Thesis: "High-resolution measurement methods for the characterization of steel in Scanning Electron Microscopy"
- Specialized in advanced lectures on special material analysis methods and modern material classes

Göttingen, Germany

Aug. 2017 - Sep. 2020

B.Sc. in Material Science

GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN

- Thesis: "Development of bio-based straps - Adaptation and modification of suitable testing methods and determination of mechanical and physical material properties"
- Developed a comprehensive understanding of material structures, material production processes, and the sustainability of material design through an interdisciplinary curriculum.

Göttingen, Germany

Aug. 2014 - Okt. 2017

Skill Summary

MATERIALS CHARACTERIZATION		EXPERIMENTAL MECHANICS & TESTING	
XRD	Texture, 2- θ Diffraction	Mechanical testing	Bending, tensile, adhesion
SEM	EBSD, EDX, t-EBSD	Development	Experimental Setups, scaling
Synchrotron	Tomography, Texture	DATA ANALYSIS & ENGINEERING TOOLS	
μ -CT	Ice samples	Data analysis	Python, Matlab, C
Opt. Microscopy	Imaging, Phase contrast	CAD	Autodesk Inventor
Spectroscopy	IR, Raman	Office	LaTeX, MS-Office

Volunteer Experience

Head Coach

TRONDHEIM LHK

- Developed and led training sessions
- Motivated and guided players

UM Fieldhockey

DOMMERKOMITEE

- Managing umpire assignment
- Facilitated the development of umpires

References

Prof. Dr. Jianying He

DEPARTMENT FOR STRUCTURAL ENGINEERING (KT), NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (NTNU)

- E-Mail: jianying.he@ntnu.no

Prof. Dr. Zhiliang Zhang

KT, NTNU

- E-Mail: zhiliang.zhang@ntnu.no

Dr. Sönke Maus

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING (IBM), NTNU

- E-Mail: sonke.maus@ntnu.no

Languages

German

Native



English

Proficient



Norwegian

Advanced



Trondheim, Norway

2017 - Present

Trondheim, Norway

2003 - Present

Trondheim, Norway

2015 - Present